

MH2500 Probability and Introduction to Statistics

Slides for Week 10-1

Topics: Expectation of sums

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Expectation of Sums of Random Variables

In the case of a single random variable (discrete or continuous), we have

$$\mathbb{E}(g(X)) = \sum_x g(x)p(x), \quad \text{or} \quad \mathbb{E}(g(X)) = \int_{\mathbb{R}} g(x)p(x)dx.$$

A similar result holds for two random variables.

Proposition

If X and Y have a joint mass/density function $p(x, y)$ then

$$\begin{aligned} \mathbb{E}(g(X, Y)) &= \sum_{x,y} g(x, y)p(x, y) \quad \text{or} \\ \mathbb{E}(g(X, Y)) &= \iint_{\mathbb{R}^2} g(x, y)p(x, y)dxdy. \end{aligned}$$

Corollary

$\mathbb{E}(X + Y) = \mathbb{E}(X) + \mathbb{E}(Y)$ whenever $\mathbb{E}(X)$ and $\mathbb{E}(Y)$ exist.

Proof: We prove the corollary in the continuous case.

$$\begin{aligned}\mathbb{E}(X + Y) &= \iint (x + y)p(x, y)dx dy \\&= \iint xp(x, y)dx dy + \iint yp(x, y)dx dy \\&= \int x \left(\int p(x, y)dy \right) dx + \int y \left(\int p(x, y)dx \right) dy \\&= \int x p_X(x)dx + \int y p_Y(y)dy \\&= \mathbb{E}(X) + \mathbb{E}(Y)\end{aligned}$$

Corollary

If $X \geq Y$ then $\mathbb{E}(X) \geq \mathbb{E}(Y)$.

Proof:

$$X - Y \geq 0 \quad \Rightarrow \quad \mathbb{E}(X - Y) \geq 0 \quad \Rightarrow \quad \mathbb{E}(X) - \mathbb{E}(Y) \geq 0.$$

Definition

Let $X_1, \dots, X_n$ be i.i.d. variables having distribution function F and expected value μ . Such a sequence of random variables is said to constitute **a sample from the distribution F** .

- The quantity $\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i$ is called the **sample mean**.
- $\mathbb{E}(\bar{X}) = \mu$, i.e., $\bar{X}$ is an unbiased estimator of μ .

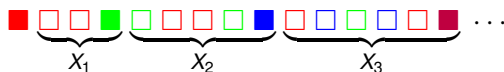
Proof:

$$\mathbb{E}(\bar{X}) = \mathbb{E}\left(\frac{1}{n} \sum_{i=1}^n X_i\right) = \frac{1}{n} \mathbb{E}\left(\sum_{i=1}^n X_i\right) = \frac{1}{n} \sum_{i=1}^n \mathbb{E}(X_i) = \mu.$$

Coupon Collector Problem

There are N types of coupons. Each time one obtains a coupon, it is equally likely to be any one of the N types. Find the expected number of coupons one needs to amass before obtaining a complete set of at least one of each type.

Solution: Let X denote the number of coupons collected before a complete set is attained.



In general, X_i = the number of additional steps needed to see the $(i + 1)$ -st type after having collected distinct i types.

$$\mathbb{E}(X) = 1 + \mathbb{E}(X_1) + \mathbb{E}(X_2) + \cdots + \mathbb{E}(X_{N-1})$$

Observe that $X_i \sim \text{Geometric}((N - i)/N)$ and thus

$$\mathbb{E}(X_i) = \frac{N}{N - i}.$$

Therefore

$$\begin{aligned}\mathbb{E}(X) &= 1 + \mathbb{E}(X_1) + \mathbb{E}(X_2) + \cdots + \mathbb{E}(X_{N-1}) \\ &= 1 + \frac{N}{N-1} + \frac{N}{N-2} + \cdots + \frac{N}{1} \\ &= N \left(1 + \frac{1}{2} + \frac{1}{3} + \cdots + \frac{1}{N} \right) \\ &\approx N \ln N.\end{aligned}$$

Recall the Euler's constant:

$$\lim_{N \rightarrow \infty} \left(1 + \frac{1}{2} + \frac{1}{3} + \cdots + \frac{1}{N} - \ln N \right).$$

Problem: Expected Number of Runs

- Consider a random sequence of n 1s and m 0s (there are $\binom{n+m}{n} = \frac{(n+m)!}{n!m!}$ possible such sequences and we pick one uniformly).
- Any consecutive string of 1s is said to constitute a run of 1s. (For example, 1110110010 has 3 runs of 1s)
- Let R be the number of runs of 1s.
- Find $\mathbb{E}(R)$.

Let I_i be the indicator variable that a run of 1s starts at the i -th position. Then

$$\mathbb{E}(R) = \sum_{i=1}^{m+n} \mathbb{E}(I_i)$$

$$\mathbb{E}(I_1) = P\{\text{a run starts at 1st position}\} = \frac{n}{n+m}.$$

$$\begin{aligned}\mathbb{E}(I_i) &= P\{\text{a run starts at } i\text{-th position}\} \\ &= P\{0 \text{ at } (i-1)\text{-st position and } 1 \text{ at } i\text{-th position}\} \\ &= \frac{m}{n+m} \cdot \frac{n}{n+m-1}\end{aligned}$$

Hence

$$\begin{aligned}\mathbb{E}(R) &= \mathbb{E}(I_1) + \mathbb{E}(I_2) + \cdots + \mathbb{E}(I_n) \\ &= \frac{n}{m+n} + (m+n-1) \frac{mn}{(n+m)(n+m-1)} \\ &= \frac{n(m+1)}{n+m}.\end{aligned}$$

Expectation of Products of Independent Random Variables

Proposition

If X and Y are **independent**, then

$$\mathbb{E}(XY) = \mathbb{E}(X) \mathbb{E}(Y),$$

provided that $\mathbb{E}(X), \mathbb{E}(Y)$ exist.

Corollary

If X and Y are **independent random variables** and g and h are fixed functions, then

$$\mathbb{E}[g(X)h(Y)] = \mathbb{E}[g(X)] \cdot \mathbb{E}[h(Y)]$$

provided that the expectations on the right-hand side exist.